



5 February 2026

Dear Chair of the SAI Search Committee,

I am writing to give my strongest possible recommendation for **Dr. Jialun Cao** for a faculty position in your **School of Artificial Intelligence at CUHK-Shenzhen**. I was her PhD advisor at the Hong Kong University of Science and Technology from 2019 to 2024 and have continued to closely follow her academic progress since her graduation, and worked as a postdoc and Research Assistant Professor.

Over the past decades, I have supervised and mentored **more than 30 PhD students and postdoctoral researchers** in software engineering, systems, and related areas. **Dr. Cao clearly ranks among the top one or two students I have ever supervised**, and she is firmly within the **top 5% of all PhD students** I have worked with in terms of research depth, originality, independence, and long-term academic potential. Very few junior researchers demonstrate this level of maturity, productivity, and vision at such an early career stage.

Dr. Cao's research lies at the intersection of **software engineering and artificial intelligence**, with a particular emphasis on **AI-assisted software development, software testing, and program verification**. These areas align closely with the Department's strengths in **software systems, artificial intelligence, trustworthy computing, and empirical software engineering**, and address foundational challenges in building reliable, secure, and verifiable AI-enabled software systems.

Her research excellence has been recognized by multiple highly selective and prestigious awards. Most notably, her doctoral dissertation received the **ACM SIGSOFT 2025 Outstanding Dissertation Award**, the highest doctoral-level honor in the global software engineering community, typically awarded to only **one recipient worldwide each year**. In addition, she received the **2025 Young Scientist Award in Engineering Science** from the **Hong Kong Institution of Science**, an award conferred on **only one recipient each year across Hong Kong**. These honors jointly reflect the originality, technical depth, and long-term significance of her research contributions.

Dr. Cao has established an outstanding publication record. During her PhD studies, she published **14 papers in top-tier venues**, including **ICSE, FSE, ASE, TOSEM, and USENIX Security**. To date, she has authored **30 peer-reviewed publications**, which have accumulated **over 700 Google Scholar citations**, with a strong and steadily increasing citation trajectory. Among the doctoral students I have supervised, **her publication quality, volume, and impact place her in the top few percent at a comparable career stage**.

One of her most influential contributions is the **DeepFD framework (ICSE 2022, JASE'24)**, which represents a major advance in automated fault localization and diagnosis for deep learning programs. This series of works address a critical challenge in maintaining large-scale AI systems and has achieved both academic and industrial impact, with **199 citations** to date and adoption by multiple industrial partners.



Beyond individual research outputs, Dr. Cao has already demonstrated a strong ability to **attract competitive research funding and sustain a long-term research program**. Her work has been supported by **competitive national and industry funding totaling HK\$5.87 million (approximately £642,000)**. This includes serving as **Co-Principal Investigator on a major Research Grants Council (RGC) of Hong Kong General Research Fund project**, as well as **Co-PI or key contributor on multiple industry-funded projects with Huawei Hong Kong**. These achievements are particularly impressive for a researcher at her career stage and clearly indicate her readiness to compete for external funding and build an independent research group at NUS.

Dr. Cao has articulated a **clear, coherent, and forward-looking research vision** that aligns strongly with NUS priorities in **AI-assisted software development, testing, formal methods, and empirical software engineering**. Her long-term goal is to develop **next-generation intelligent and verifiable software development pipelines**, integrating requirement analysis, architecture design, automated implementation, testing, and formal verification. This vision positions her well to contribute not only to existing departmental strengths, but also to emerging research capacity that supports the Department's long-term strategic ambitions.

Importantly, Dr. Cao demonstrated **research independence at an unusually early stage**. By the latter part of her PhD, she was already independently identifying research problems, leading projects, securing funding, and mentoring junior students. In terms of independence, leadership, and research judgment, **she was operating at a level comparable to a strong researcher rather than a typical PhD student**.

Dr. Cao is also an active and highly respected contributor to the academic community. She serves on the **program committees of top-tier venues**, including ICSE, FSE, ASE, ISSTA, and others. She has served as **Session Chair at ESEC/FSE 2025 and ASE 2024**, and as **Local Chair for the 2025 Frontiers Forum on AI and Formal Methods**, which featured invited speakers including a **Turing Award laureate**. In addition, she was invited to serve on the ACM SIGSOFT Outstanding Research Award 2025, Distinguished Service Award 2025, and Influential Educator Award 2025 committees, an honor rarely extended to researchers so early in their careers.

Dr. Cao has been invited to give academic talks at **University College London, Peking University, Tsinghua University, Zhejiang University**, as well as **CCF seminars and industry forums**, demonstrating her growing international visibility and leadership within the research community. She is deeply committed to fostering collaboration, supporting junior researchers, and strengthening ties between academia and industry, values that align closely with NUS's mission.

In teaching and mentorship, Dr. Cao is equally impressive. She has served as the **instructor** for undergraduate programming courses, including COMP1021 Introduction to Computer Science (with approximately **110 undergraduate students**), where she received **significantly above-average teaching evaluations**. She has also as a teaching assistant for multiple software engineering and Java programming courses, and actively co-supervises and mentors PhD students from multiple institutions. Her



teaching is research-driven, inclusive, and highly effective across both undergraduate and postgraduate levels.

Beyond research and teaching, Dr. Cao is deeply committed to **diversity, equity, and inclusion (EDI)** within the computing community. As a female scholar in software engineering, she actively advocates for gender equity and was invited to speak at the **inaugural Forum for Women Scholars in Software Engineering at CCF ChinaSoft**, encouraging young women to pursue careers in technology. She was also shortlisted as the **sole HKUST participant** for the **2024 Rising Stars Women in Engineering Workshop**.

She further strengthened her leadership and inclusion capacity by participating in the 5th Cohort of the Asia-Pacific Women in Leadership Program (2024-2025), mentored by Professor Paola Ricaurte Quijano (Harvard University and Tecnológico de Monterrey). Her commitment to inclusion extends to mentoring students from diverse cultural and religious backgrounds, particularly international students facing academic or personal challenges, helping them build confidence, research experience, and successful academic pathways.

In summary, **Dr. Jialun Cao is one of the very best doctoral students I have supervised in my academic career**, and I consider her to be **fully ready for a faculty position in your School of Artificial Intelligence at CUHK-Shenzhen**. She possesses the research independence, funding capability, teaching excellence, leadership, and long-term vision required to build a strong and internationally competitive research program.

I recommend Dr. Jialun Cao without reservation and give her my highest endorsement. Please do not hesitate to contact me should you require any additional information.

Sincerely,

Shing-Chi Cheung
Chair Professor of Computer Science and Engineering, IEEE Fellow